

FARRUKH HUSSAIN

Python Software Engineer | AI/ML | Backend Development

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PROFESSIONAL SUMMARY

Python Software Engineer with 7+ years of experience building scalable distributed systems, AI/ML solutions, and high-performance backend applications using Python ecosystem. Deep expertise in Python frameworks (Django, Flask, FastAPI), machine learning libraries (TensorFlow, PyTorch, Scikit-learn), and data processing tools with proven track record of delivering production systems serving millions of users. Specialized in generative AI, natural language processing and Python-based Microservices architecture with hands-on experience in cloud platforms. Strong background in Python system design, algorithms, and DevOps/MLOps practices.

TECHNICAL SKILLS

Programming Languages: Python (Expert), SQL, JavaScript, Bash, Java

Python Frameworks & Libraries: Django, Flask, FastAPI, Tornado, Pyramid, SQLAlchemy, Celery, Pandas, NumPy, SciPy

Python AI/ML Libraries: TensorFlow, PyTorch, Scikit-learn, Keras, OpenAI's API, LangChain, LlamaIndex, Hugging Face Transformers, NLTK, SpaCy

Python Data Tools: Pandas, NumPy, Dask, Polars, Matplotlib, Seaborn, Jupyter, Apache Airflow

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Vector Databases (Pinecone, FAISS)

Cloud & Infrastructure: AWS (EC2, S3, Lambda, SageMaker, RDS), GCP (Vertex AI), Azure, Docker, Kubernetes

Python DevOps Tools: Docker, Kubernetes, Jenkins, pytest, tox, black, flake8, mypy

Big Data & Streaming: Apache Spark (PySpark), Kafka, Hadoop, ETL Pipelines

PROFESSIONAL EXPERIENCE

Senior Machine Learning Engineer | Ambertek

May 2023 - Present · 2 yrs 5 mos

- Designed and developed Python Microservices architecture serving 1M+ daily users using Django, FastAPI, and asyncio, improving system throughput by 40%.
- Built end-to-end ML pipelines using Python with TensorFlow, PyTorch, and Hugging Face Transformers, reducing model inference latency from 500ms to 120ms.
- Implemented distributed training systems for large language models using PySpark and Python multiprocessing, scaling to process 10TB+ datasets daily.
- Developed real-time anomaly detection system using Python scikit-learn and custom algorithms (Isolation Forest, Autoencoders), achieving 99.2% accuracy in production.
- Architected cloud-native Python applications on AWS using Lambda, EC2, and SageMaker with auto-scaling capabilities, reducing infrastructure costs by 35%.
- Established Python-based MLOps workflows using MLflow, Airflow, and automated testing with pytest, decreasing deployment time from 4 hours to 15 minutes.
- Optimized Python neural network architectures through hyperparameter tuning with Optuna and Bayesian optimization, improving model performance by 25%.
- Led cross-functional team of 5 Python developers in agile development, delivering 15+ production features using Python best practices.

Senior Software Engineer | SlashNext

Aug 2022 – May 2023

- Architected distributed Python systems for training and deploying generative AI models (GPT, BERT) using PyTorch and Python multiprocessing, supporting concurrent model training across 50+ GPU instances.
- Built transformer-based architectures using Python and Hugging Face for text generation, summarization, and semantic search, achieving BLEU scores >0.85 for production models.
- Developed multimodal AI systems using Python, integrating text, image, and structured data with CLIP and DALL-E APIs, processing 100K+ documents daily.
- Engineered high-performance data processing pipelines using Pandas, Dask, and Polars in Python, reducing data preprocessing time by 60%.
- Created RESTful APIs and GraphQL endpoints using FastAPI and Flask, handling 10K+ requests per second with 99.9% uptime and Python async/await patterns.

- Implemented vector search engines using Python with FAISS and Pinecone for Retrieval-Augmented Generation (RAG) workflows, improving search relevance by 45%.
- Deployed containerized Python applications using Docker and Kubernetes on AWS EKS, achieving zero-downtime deployments with Python health checks.
- Integrated document processing solutions using Python libraries (OCR, Tesseract, PyPDF2) and computer vision models for unstructured data extraction.

AI Engineer | Nelda Group AG · Contract

Jan 2022 - Aug 2022 · 8 mos

- Led development of high-traffic e-commerce platform using Python Flask, Django ORM, and MySQL, handling 500K+ transactions daily with 99.95% availability.
- Designed and implemented distributed caching layer using Python Redis client and Elasticsearch-py, reducing page load times by 70% and improving search response times by 80%.
- Built payment processing microservices using Python with Stripe and PayPal APIs, processing \$50M+ annually with PCI DSS compliance and comprehensive error handling.
- Developed recommendation engine using Python scikit-learn, collaborative filtering, and TensorFlow deep learning models, increasing user engagement by 35%.
- Implemented A/B testing framework using Python statistics libraries and real-time analytics with PySpark and Kafka-python, enabling data-driven product decisions.
- Established CI/CD pipelines using Python scripts, Jenkins, and GitLab with automated testing using pytest, reducing deployment failures by 90% and enabling daily releases.
- Fine-tuned GPT-2 models using Python transformers library for automated content generation, implementing transfer learning techniques with Java integration for legacy systems.
- Mentored 3 junior Python developers and conducted technical interviews focusing on Python algorithms and system design.

Machine Learning Engineer | Afiniti,

Jul 2018 – Apr 2022

- Developed scalable Python backend systems using Django, Flask, and asyncio for data-intensive applications with multi-threading and async processing.
- Designed RESTful APIs using Python frameworks and microservices architecture, improving system modularity and enabling independent deployments with containerization.
- Built automated ETL pipelines using Pandas, NumPy, and SQLAlchemy with Python scripts, processing 1TB+ data daily with 99.5% data quality and error handling.

- Implemented message queuing systems using Celery, RabbitMQ, and Python multiprocessing for high-throughput task processing (50K+ jobs/hour). Integrated AWS services using boto3 Python SDK (Lambda, S3, EC2, RDS) for serverless computing and auto-scaling applications.
- Optimized Python database queries using SQLAlchemy ORM and implemented connection pooling with Java database connectors, reducing query response time by 50%.

AI Research Associate | Information Technology University

Mar 2017 – Apr 2018

- Detected online scam crypto pages using machine learning
- Designed a classifier for malicious documents detection using machine learning and deep learning
- Data Annotation. Data Augmentation. Production Monitoring.
- Worked on cross-region building counting from satellite imagery.
- To minimize the domain gap, we employed the following techniques:
- Ranking within an Image
- Ranking across Images
- Output Space Adaptation

EDUCATION

Bachelor of Science in Information Technology & Systems

Preston University, Islamabad

LinkedIn

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